# Amith Sirisilla Jayadevgari

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## PROFESSIONAL SUMMARY

Computer Science student specializing in deploying local Large Language Models (LLMs) and building high-performance, spatial-temporal Machine Learning (ML) pipelines. Proven track record engineering zero-idle-memory RAG pipelines, vision-language OCR parsers, and real-time spatial database compute engines with sub-second query latency. Adept at optimizing inference pipelines, vector databases, and 3D WebGL visualizations using PyTorch, DuckDB Spatial, Jina Embeddings, and Apache Arrow.

## KEY PROJECTS

**MedRAG (Clinical RAG Pipeline)** | GitHub | *Python, LanceDB, Qwen 3.5, Chandra OCR 2, Jina Embeddings v5, FastAPI*

- Reduced query latency in a Natural Language Processing (NLP) pipeline to **¡150ms** by implementing semantic search and indexing on disk via the **LanceDB** vector database, maintaining a **0MB idle RAM footprint**.
- Achieved **94%** parsing accuracy on layout-dense clinical datasets by deploying the **Chandra OCR 2** vision-language model, generating chunk-free vector embeddings via **Jina Embeddings v5** (32k context window).
- Sustained offline model inference speeds of **14 tokens/sec** by applying 5-bit integer quantization (Q5_K_M) parameters to a local **Qwen 3.5 9B** LLM, optimizing memory baseline targets to **16GB** on Apple Silicon.

**CityPlanner (Urban Optimization Agent)** | GitHub | *Python, TypeScript, Gemma AI, DuckDB Spatial, Apache Arrow, FastAPI, Deck*

- Automated multi-objective zoning planning by designing an AI agent system utilizing a local **Gemma** LLM for natural-language prompt processing and spatial feature engineering, cutting iterations by **65%**.
- Accelerated spatial database queries and relational vector intersections to **50k+ operations/sec** by designing a spatial indexing compute engine in **DuckDB Spatial**, cutting pipeline runtimes by **40%**.
- Eliminated dataset serialization latency by **70%** during real-time data ingestion, streaming structured tables via **Apache Arrow IPC** to a 3D WebGL Deck.gl dashboard rendering at **60 FPS**.

**Project PEANUT (Environmental Forecasting)** | GitHub | *TypeScript, Python, Next.js, GEE, Scikit-Learn, Tailwind, Supabase*

- Processed over **120GB of multispectral satellite imagery** via the **Google Earth Engine (GEE) API** for feature extraction of vegetation indices, scaling data pipeline analysis coverage by **3.5x**.
- Trained a multi-variable Polynomial Regression model in **Scikit-Learn** to predict resource deficits, achieving an **$R^2$ score of 0.88** and optimizing predictive accuracy over population projection datasets.
- Designed a spatial mapping dashboard serving predictions from supervised classification models, streaming structured GeoJSON layers via custom REST APIs.

## TECHNICAL SKILLS

**Programming Languages**: Python, TypeScript, JavaScript, SQL, C, Bash
**Frameworks & APIs**: React, Next.js, Node.js, FastAPI, Streamlit, Deck.gl, Maplibre GL, Tailwind CSS
**Machine Learning & Deep Learning**: PyTorch, TensorFlow, Scikit-Learn, Deep Learning Models, Hugging Face, Jina Embeddings, Ollama, LangChain, LlamaIndex, Retrieval-Augmented Generation (RAG), Semantic Search
**NLP & Computer Vision**: CNNs, LSTMs, YOLOv8, OpenCV, Visual OCR, Explainable AI (XAI), Grad-CAM
**Databases & Spatial Computing**: Google Earth Engine (GEE) API, DuckDB (Spatial Extension), Apache Arrow (IPC / PyArrow), LanceDB (Vector DB), Supabase, PostgreSQL
**Tools & Methodologies**: Git, GitHub, Docker, Jupyter Notebook, Linux Systems, Agile/Scrum, OOP

## CERTIFICATIONS

**Google Data Analytics Professional Certificate** | *Coursera*
**IBM Machine Learning Professional Certificate** | *Coursera*
**IBM Relational Database Administration (DBA)** | *Coursera*
**Artificial Intelligence Foundations: Thinking Machines** | *LinkedIn Learning*

## EDUCATION

**Lovely Professional University** — Jalandhar, Punjab
*Bachelor of Technology in Computer Science and Engineering; CGPA: 7.0/10.0* — *2022 – 2026*

## LEADERSHIP & ACHIEVEMENTS

- **NGO Leadership & Volunteering:** Coordinated a team of 15 volunteers to execute outreach campaigns for child education, directly teaching and supporting 150+ underprivileged children.
- **Community Organizing:** Led local public safety and child labor awareness initiatives, coordinating with community stakeholders to organize webinars with 300+ attendees.